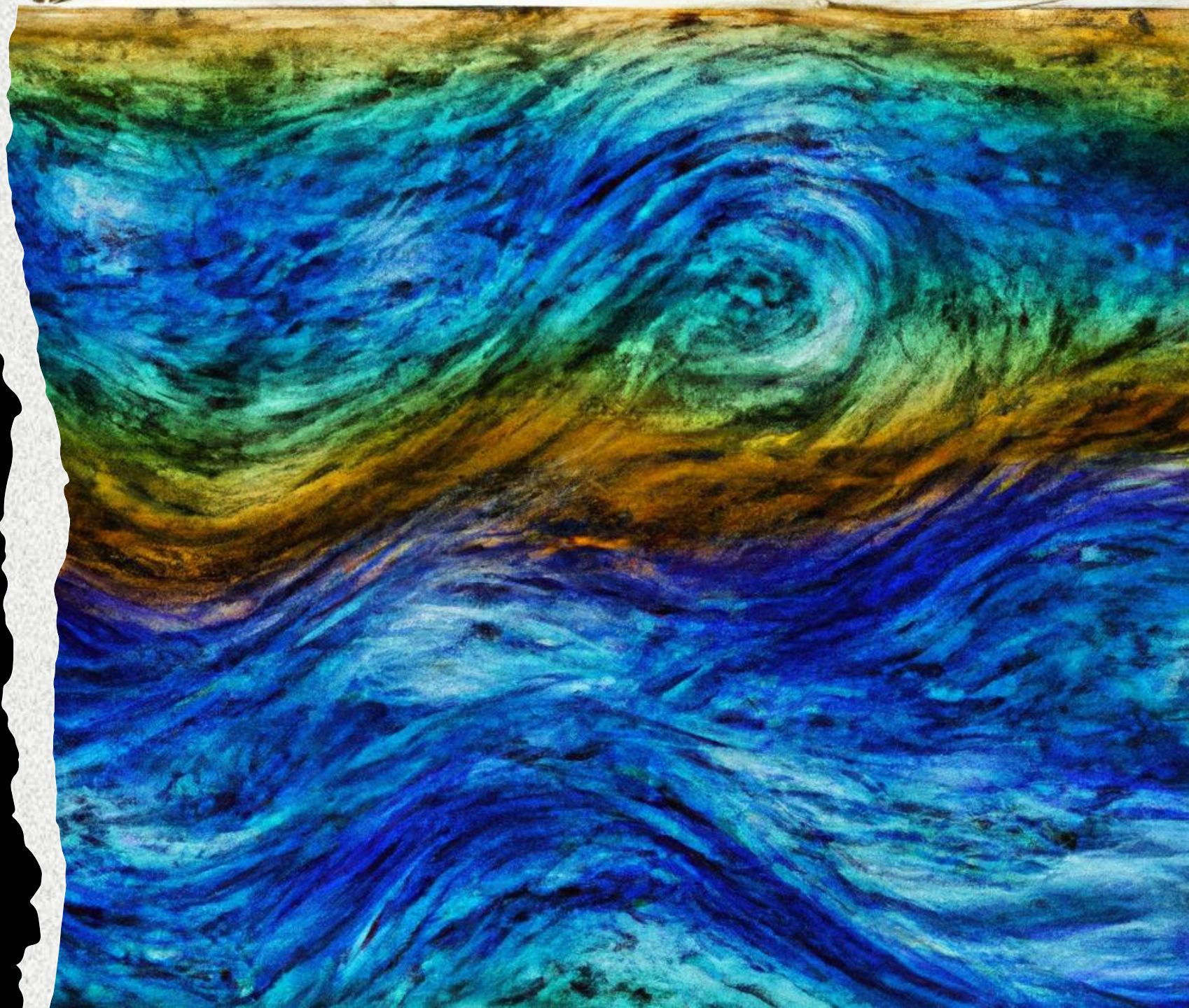
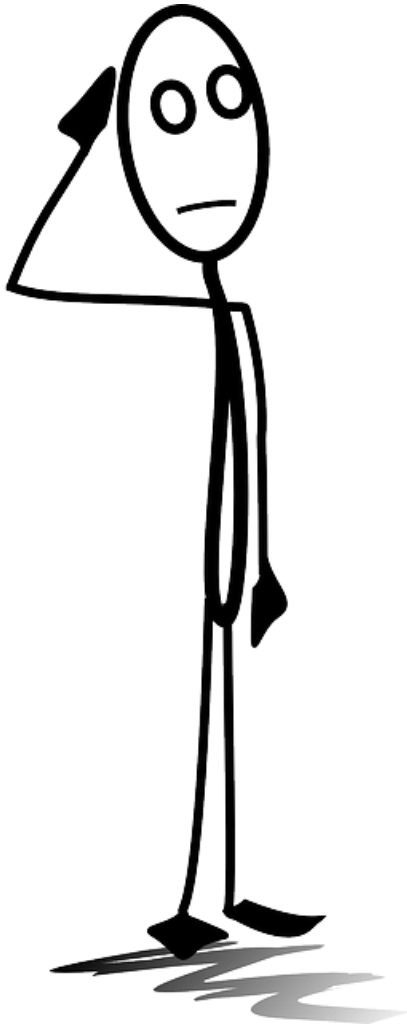
PHYSICS I
WEEK 8

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Office: HSB 116B

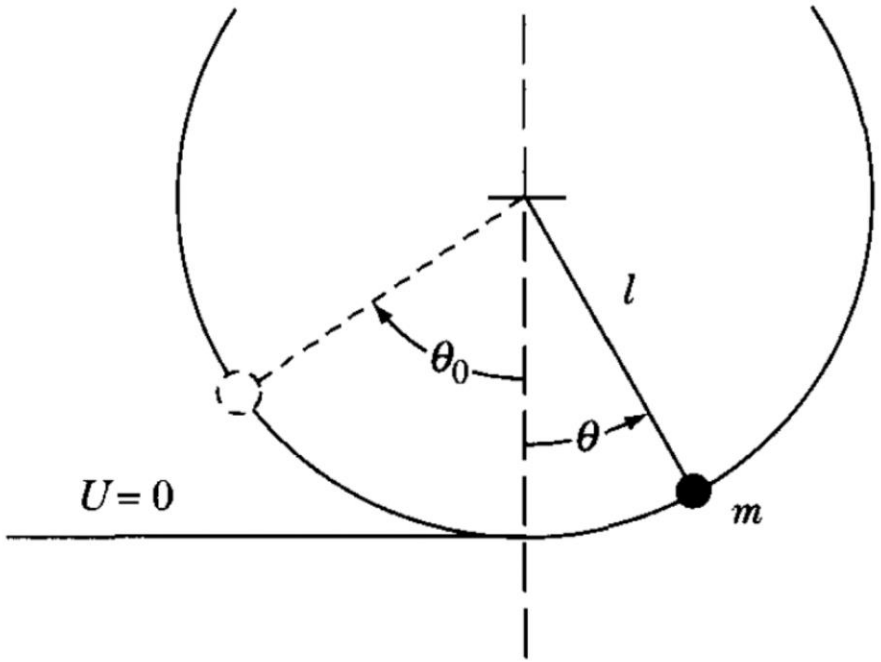


WEET 8: OUTLINE

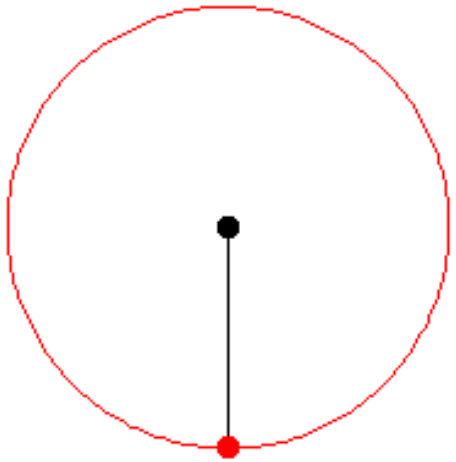
- Review: Phase portraits
- Conservative forces, gradient, curl (Taylor 4.1-4.4)
- Spherical and cylindrical coordinate systems (Taylor 4.8)

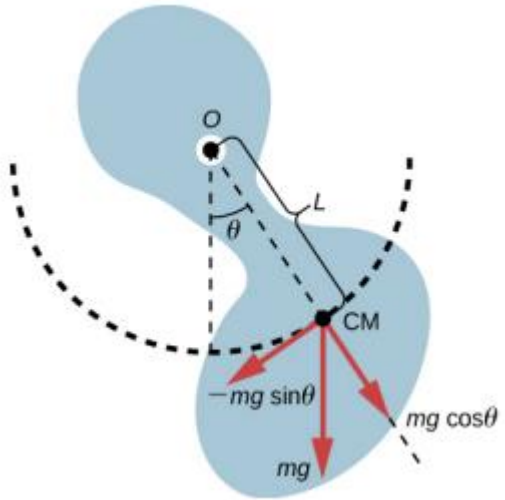


NONLINEAR SYSTEM: PLANE PENDULUM

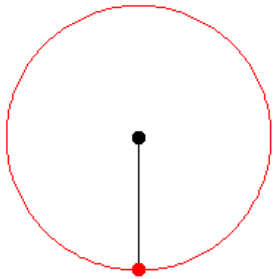


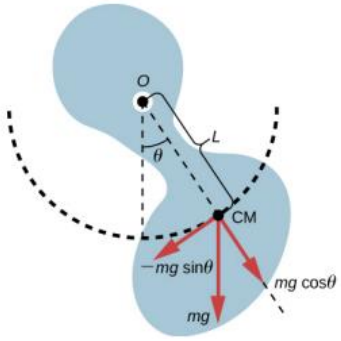
The plane pendulum where the mass m is not required to oscillate in small angles. The angle $\theta > 0$ is in the counterclockwise direction so that $\theta_0 < 0$.



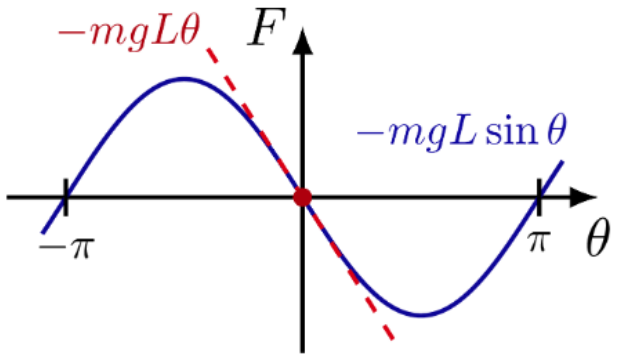


NONLINEAR SYSTEM: PLANE PENDULUM





Approximate the gravitational force on a pendulum with a Hooke's law, which is equivalent to the small-angle approximation:

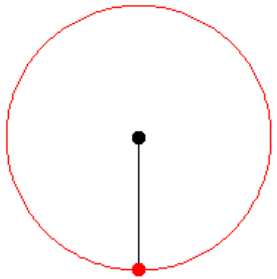


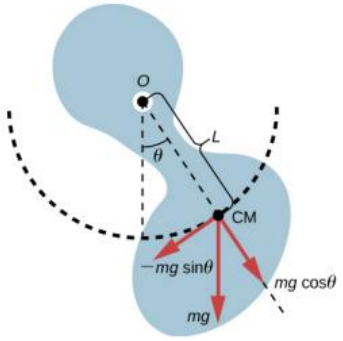
Small angle approximation

$$\ddot{\theta} + \omega_0^2 \theta = 0$$

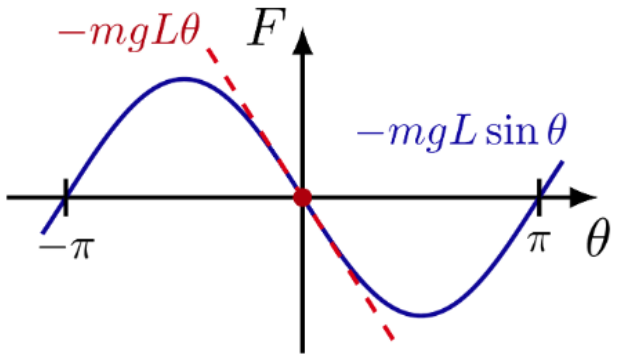
$$\tau \cong 2\pi\sqrt{\frac{l}{g}}$$

NONLINEAR SYSTEM: PLANE PENDULUM





Approximate the gravitational force on a pendulum with a Hooke's law, which is equivalent to the small-angle approximation:



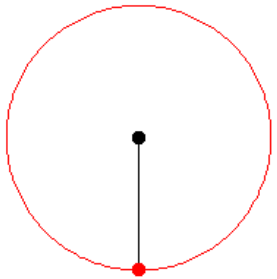
For a parabolic potential

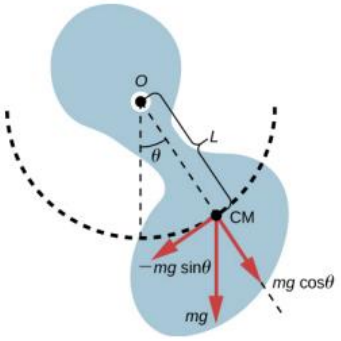
$$\frac{1}{2}m\,\dot{x}^2 + U(x) = E$$

$$\dot{x}(x) = \pm\sqrt{\frac{2}{m}}\sqrt{E - U(x)}$$

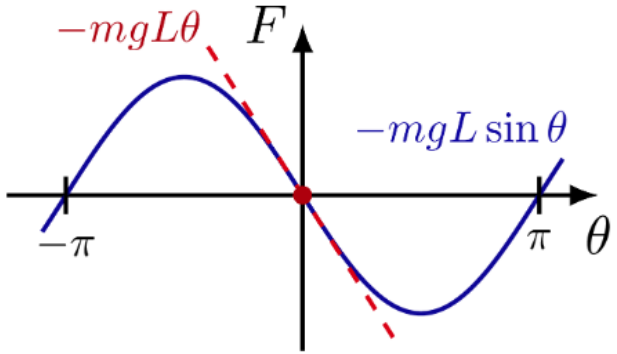
$$t = \int_{x_o}^{x} \frac{dx'}{\dot{x}(x')} = \sqrt{\frac{m}{2}}\int_{x_o}^{x} \frac{dx'}{\sqrt{E - U(x')}}$$

NONLINEAR SYSTEM: PLANE PENDULUM

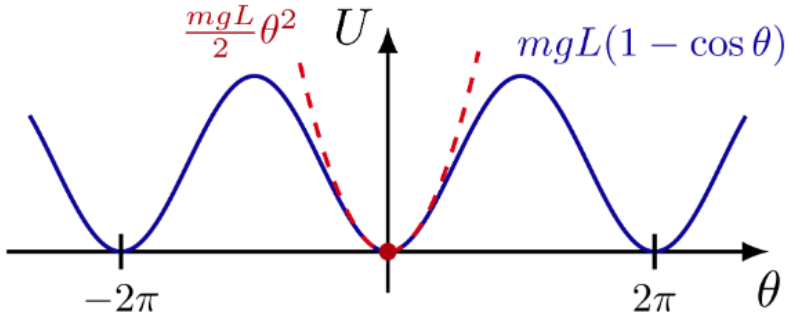




Approximate the gravitational force on a pendulum with a Hooke's law, which is equivalent to the small-angle approximation:



Approximate the potential energy of a pendulum with a parabola to get a simple harmonic oscillator:



For a parabolic potential

$$\frac{1}{2}m\,\dot{x}^2 + U(x) = E$$

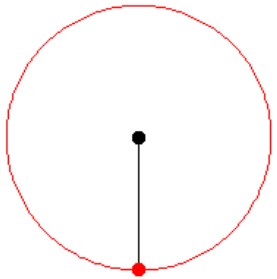
$$\Downarrow$$

$$\dot{x}(x) = \pm\sqrt{\frac{2}{m}}\sqrt{E - U(x)}$$

$$\Downarrow$$

$$t = \int_{x_0}^{x} \frac{dx'}{\dot{x}(x')} = \sqrt{\frac{m}{2}}\int_{x_0}^{x} \frac{dx'}{\sqrt{E - U(x')}}$$

NONLINEAR SYSTEM: PLANE PENDULUM



Thornton-Marion Appendix B.2

Transform the elliptic integral

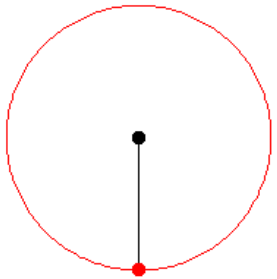
$$\int_0^{\phi} \frac{d\theta}{\sqrt{1 - n^2 \sin^2 \theta}}, \quad \text{where } n^2 > 1$$

into a standard form.

Solution. To reduce this integral to standard form, the radical must be transformed to $\sqrt{1 - k^2 \sin^2 \theta}$, with $k^2 < 1$. To do this, consider the transformation $n \sin \theta = \sin \beta$.

$$\tau \cong 2\pi \sqrt{\frac{l}{g}} \left[1 + \frac{1}{16} \theta_0^2 + \frac{11}{3072} \theta_0^4 \right]$$

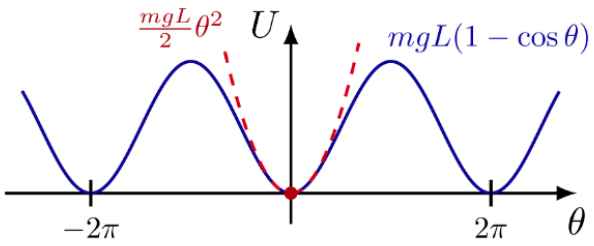
NONLINEAR SYSTEM: PLANE PENDULUM

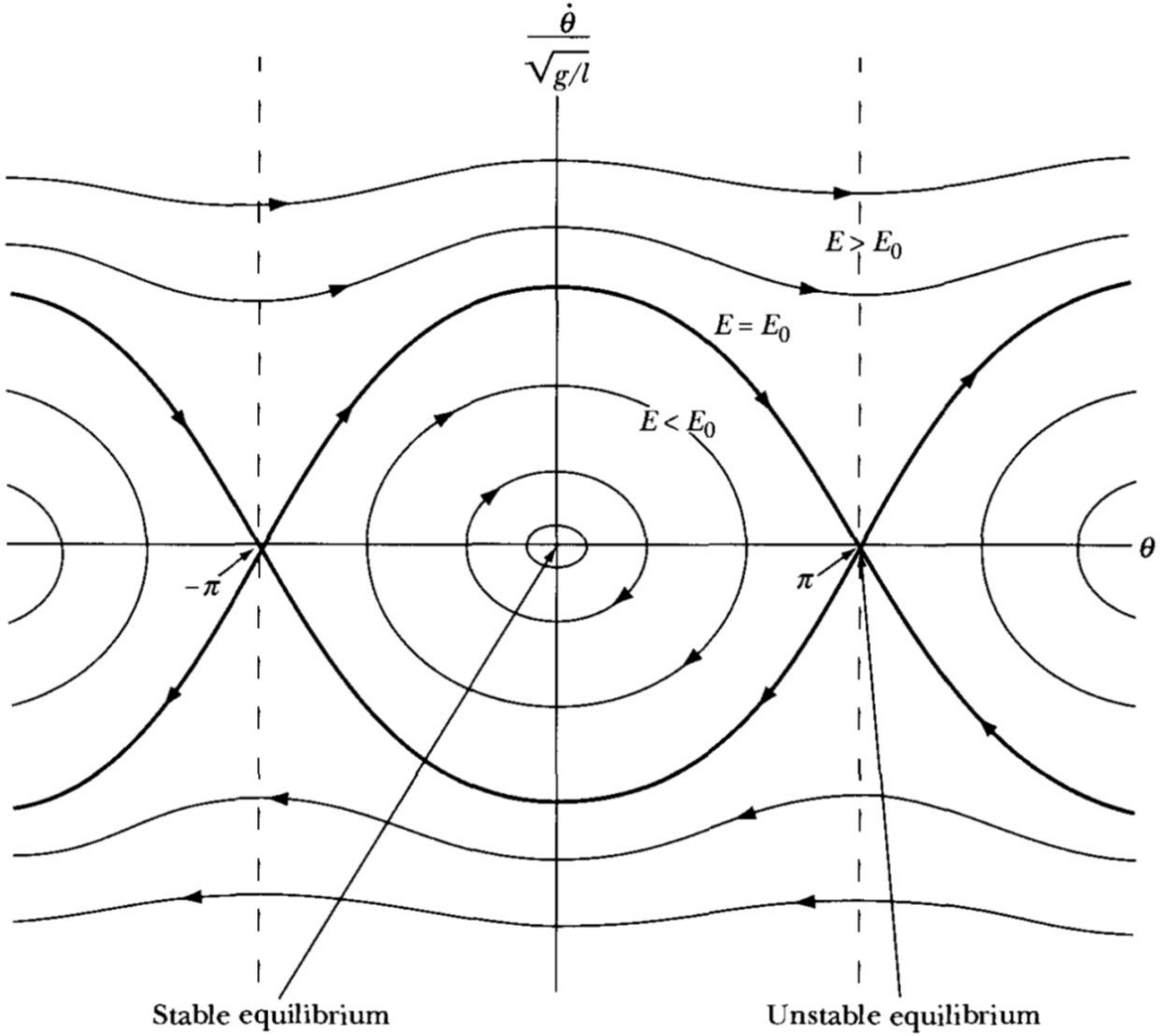


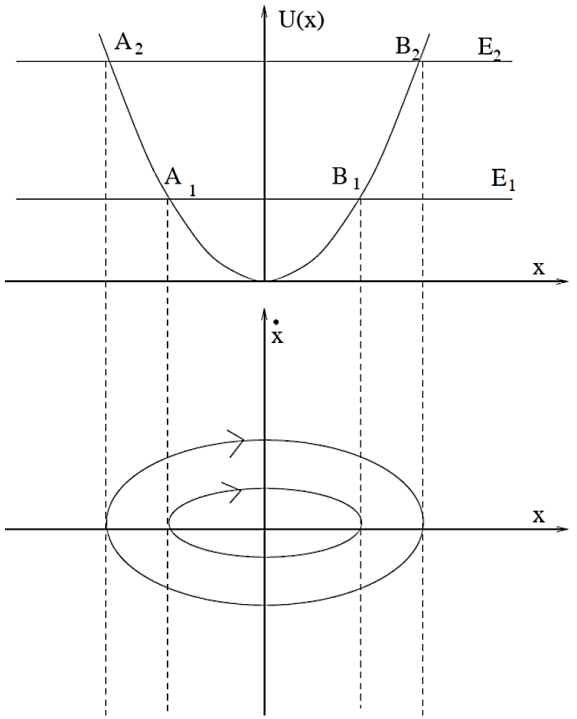
If θ and θ_0 are small

$$\left(\sqrt{\frac{l}{g}}\dot{\theta}\right)^2 + \theta^2 \cong \theta_0^2$$

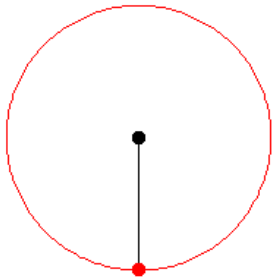
$$E < 2mgl \equiv E_0$$







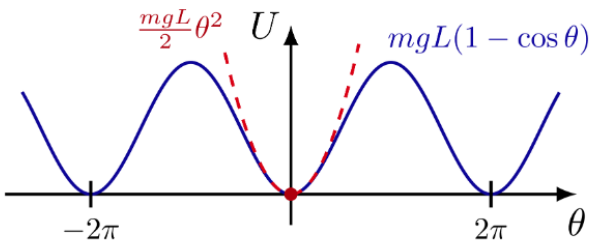
NONLINEAR SYSTEM: PLANE PENDULUM

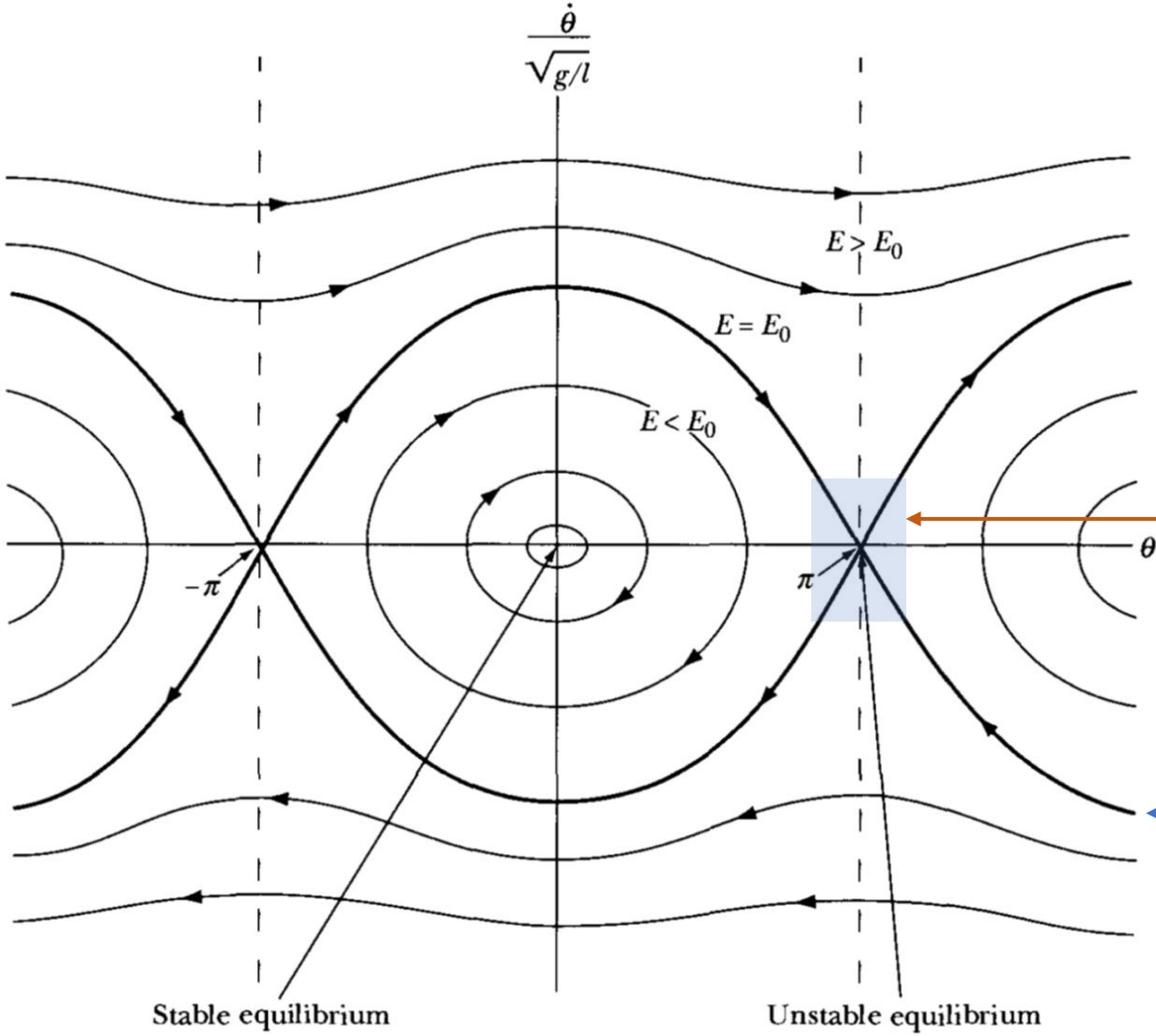


If θ and θ_0 are small

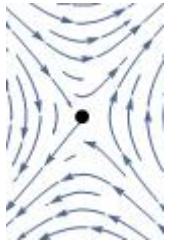
$$\left(\sqrt{\frac{l}{g}}\dot{\theta}\right)^2 + \theta^2 \cong \theta_0^2$$

$$E < 2mgl \equiv E_0$$





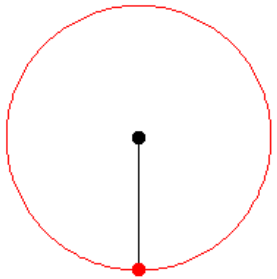
Extremely sensitive to the initial conditions



$$E = E_0$$

$$\dot{\theta} = \pm 2\sqrt{\frac{g}{l}}\cos(\theta/2)$$

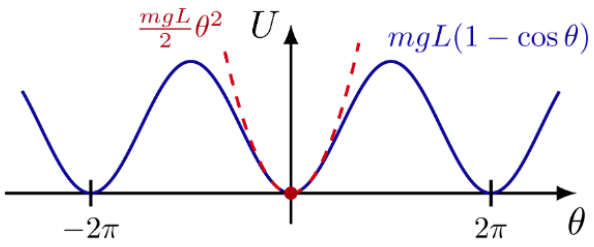
NONLINEAR SYSTEM: PLANE PENDULUM

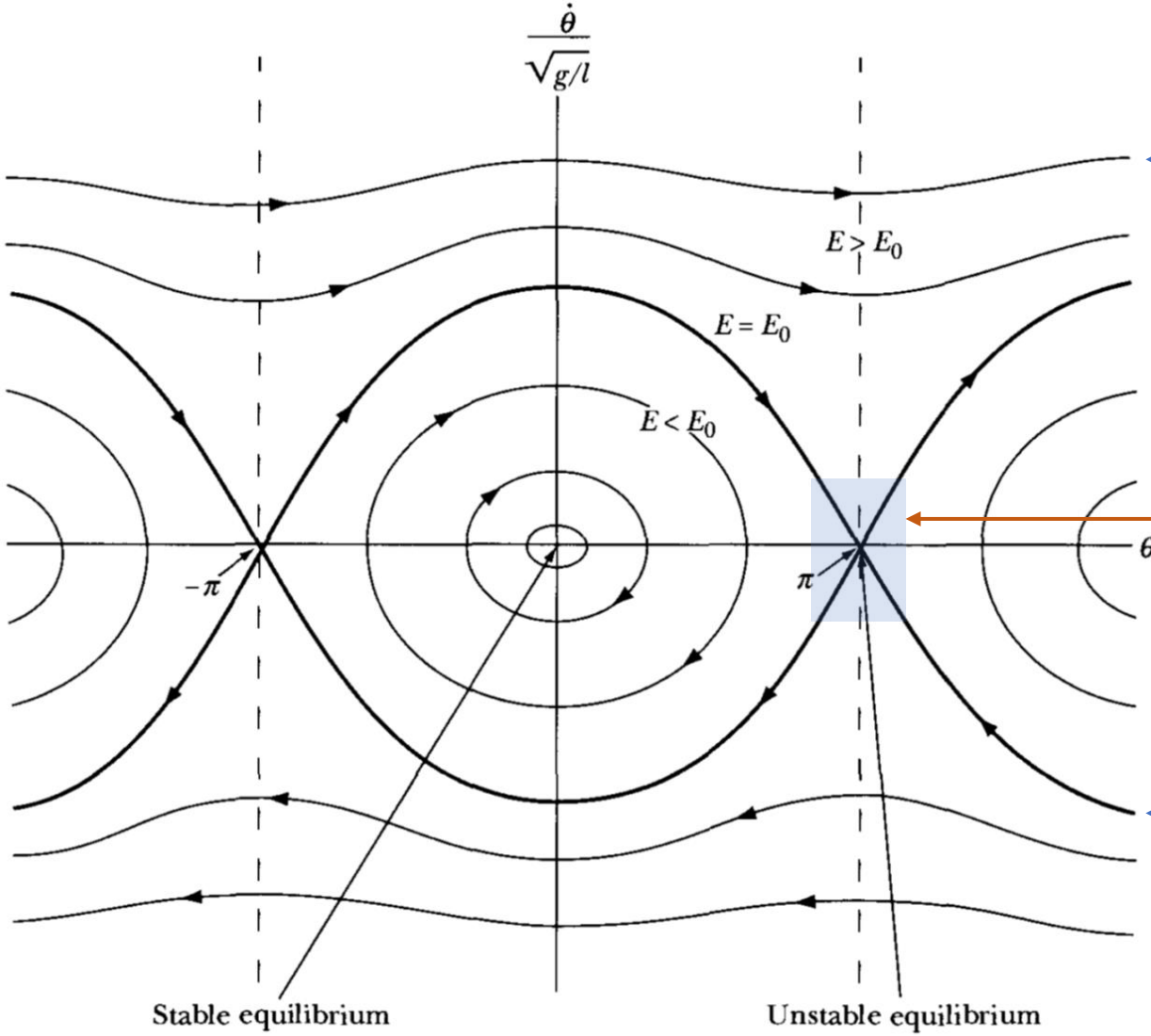


If θ and θ_0 are small

$$\left(\sqrt{\frac{l}{g}}\dot{\theta}\right)^2 + \theta^2 \cong \theta_0^2$$

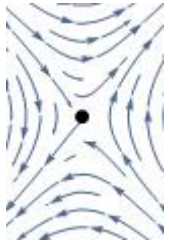
$$E < 2mgl \equiv E_0$$





No oscillations, complete revolution

Extremely sensitive to the initial conditions

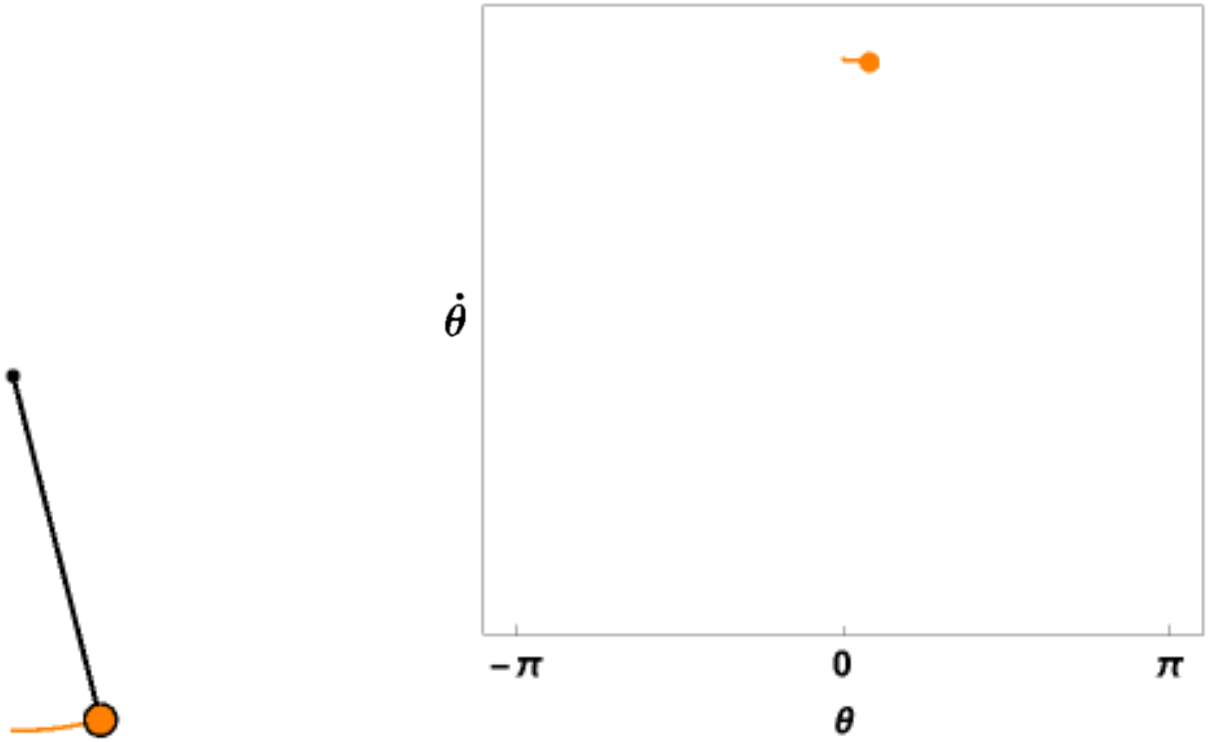


$$E = E_0$$

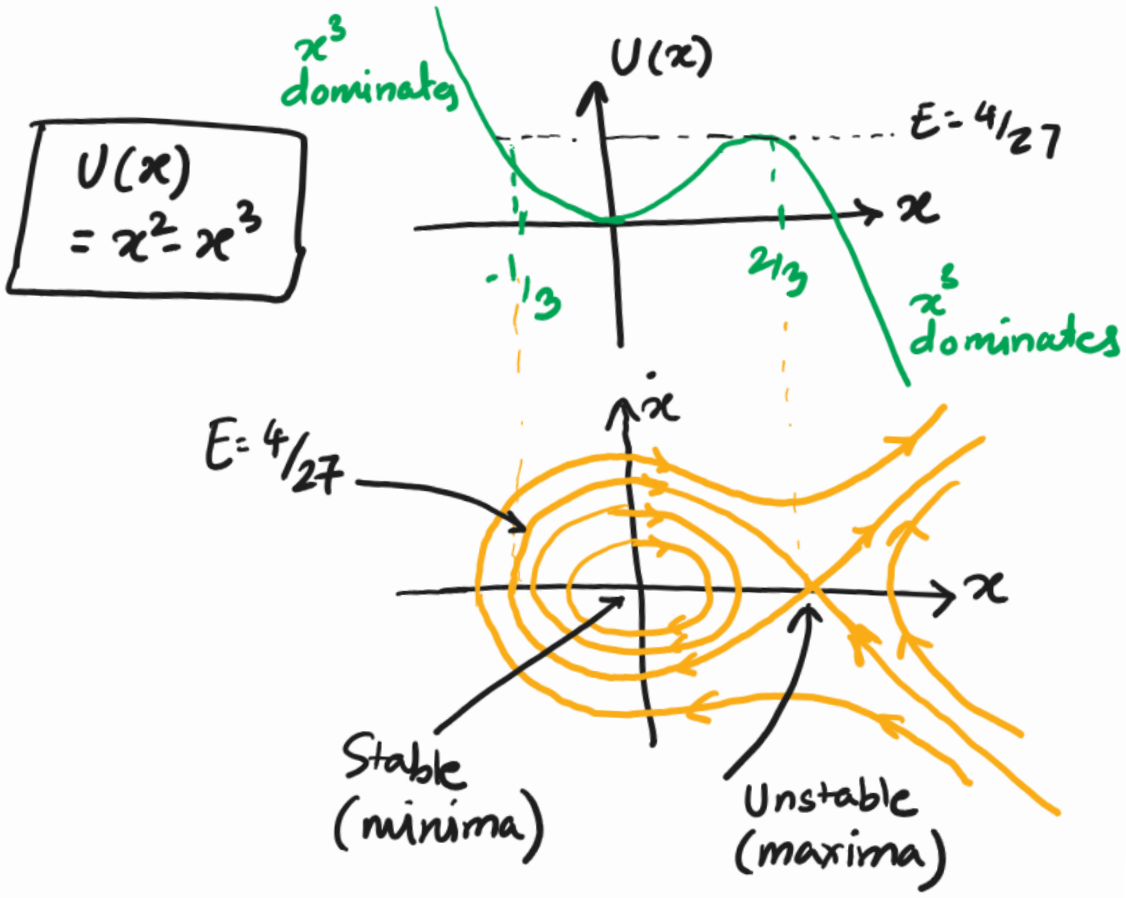
$$\dot{\theta} = \pm 2\sqrt{\frac{g}{l}}\cos(\theta/2)$$

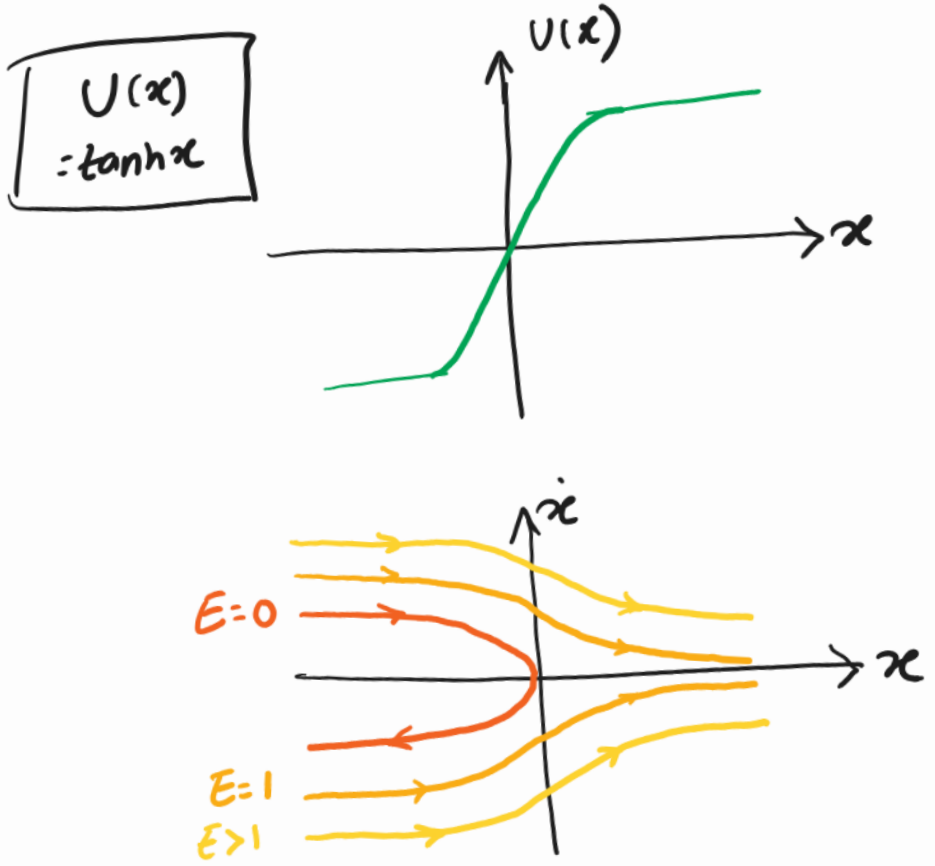
NONLINEAR SYSTEM: PLANE PENDULUM

Damped

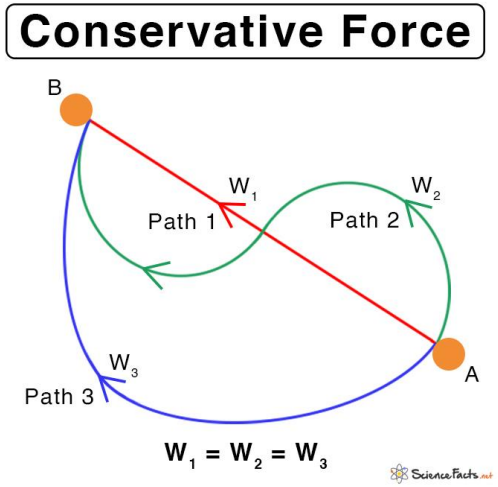


REVIEW: TUTORIAL 5





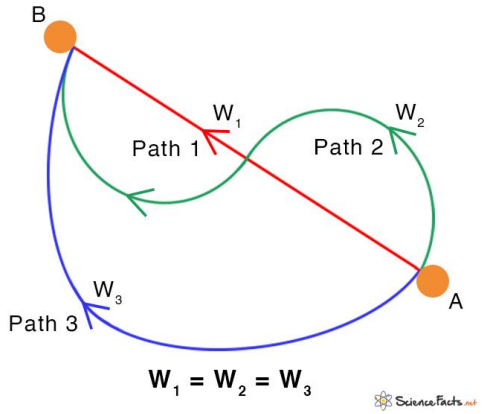
WORK-ENERGY THEOREM IN 3D



$\int_1^2 \mathbf{F} \cdot d\mathbf{r}$ is path-independent

WORK-ENERGY THEOREM IN 3D

Conservative Force



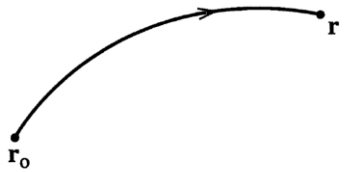
$\int_1^2 \mathbf{F} \cdot d\mathbf{r}$ is path-independent

In 1D

$$F(x) = -\frac{dU}{dx}$$

3D?

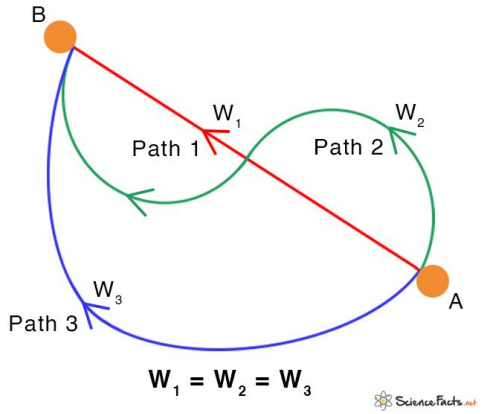
Potential Energy



$$U(\mathbf{r}) = -W(\mathbf{r}_o \to \mathbf{r}) \equiv -\int_{\mathbf{r}_o}^{\mathbf{r}} \mathbf{F}(\mathbf{r}') \cdot d\mathbf{r}'$$

WORK-ENERGY THEOREM IN 3D

Conservative Force

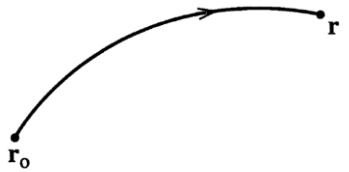


$\int_1^2 \mathbf{F} \cdot d\mathbf{r}$ is path-independent

In 1D

$$F(x) = -\frac{dU}{dx}$$

Potential Energy



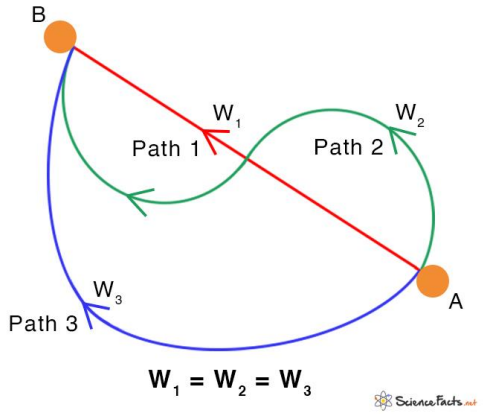
$$U(\mathbf{r}) = -W(\mathbf{r}_o \to \mathbf{r}) \equiv -\int_{\mathbf{r}_o}^{\mathbf{r}} \mathbf{F}(\mathbf{r}') \cdot d\mathbf{r}'$$

3D?

$$\mathbf{F} = -\hat{\mathbf{x}}\frac{\partial U}{\partial x} - \hat{\mathbf{y}}\frac{\partial U}{\partial y} - \hat{\mathbf{z}}\frac{\partial U}{\partial z}$$

WORK-ENERGY THEOREM IN 3D

Conservative Force

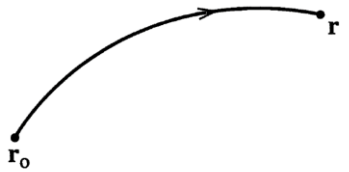


$\int_1^2 \mathbf{F} \cdot d\mathbf{r}$ is path-independent

In 1D

$$F(x) = -\frac{dU}{dx}$$

Potential Energy



$$U(\mathbf{r}) = -W(\mathbf{r}_o \to \mathbf{r}) \equiv -\int_{\mathbf{r}_o}^{\mathbf{r}} \mathbf{F}(\mathbf{r}') \cdot d\mathbf{r}'$$

3D?

$$\mathbf{F} = -\hat{\mathbf{x}}\frac{\partial U}{\partial x} - \hat{\mathbf{y}}\frac{\partial U}{\partial y} - \hat{\mathbf{z}}\frac{\partial U}{\partial z}$$

$$\boldsymbol{F} = -\boldsymbol{\nabla} U$$

∇ OPERATOR

- ∇ Operator: A vector operator

$$\nabla = \hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{z}}\frac{\partial}{\partial z}.$$

- When acts on a scalar – ∇T: **Gradient**
- When acts on a vector via the dot product – $\nabla . \boldsymbol{T}$: **Divergence**
- When acts on a vector via the cross product– $\nabla \times \boldsymbol{T}$: **Curl**

∇ OPERATOR

- Gradient

$$\nabla T \equiv \frac{\partial T}{\partial x}\hat{\mathbf{x}} + \frac{\partial T}{\partial y}\hat{\mathbf{y}} + \frac{\partial T}{\partial z}\hat{\mathbf{z}}$$

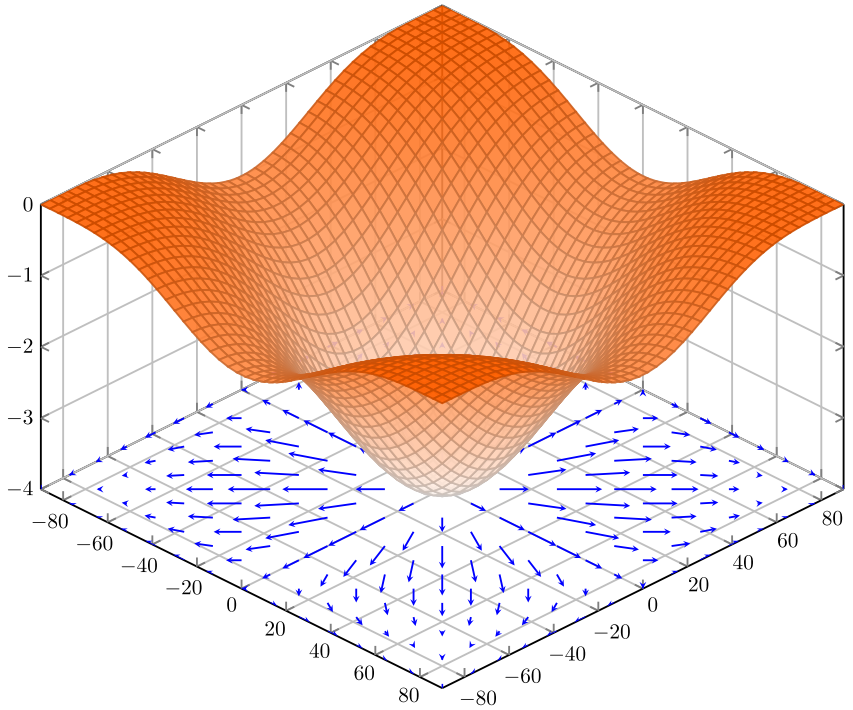
- The magnitude $|\nabla T|$ gives the slope (rate of increase) along the maximal direction
- The gradient ∇T points in the direction of maximum increase of the function T

∇ OPERATOR

- Gradient

$$\nabla T \equiv \frac{\partial T}{\partial x}\hat{\mathbf{x}} + \frac{\partial T}{\partial y}\hat{\mathbf{y}} + \frac{\partial T}{\partial z}\hat{\mathbf{z}}$$

- The magnitude $|\nabla T|$ gives the slope (rate of increase) along the maximal direction
- The gradient ∇T points in the direction of maximum increase of the function T



A 2D function

WORK-ENERGY THEOREM IN 3D

The potential energy of a certain particle is $U = Axy^2 + B \sin Cz$, where A, B and C are constants. What is the corresponding force?

WORK-ENERGY THEOREM IN 3D

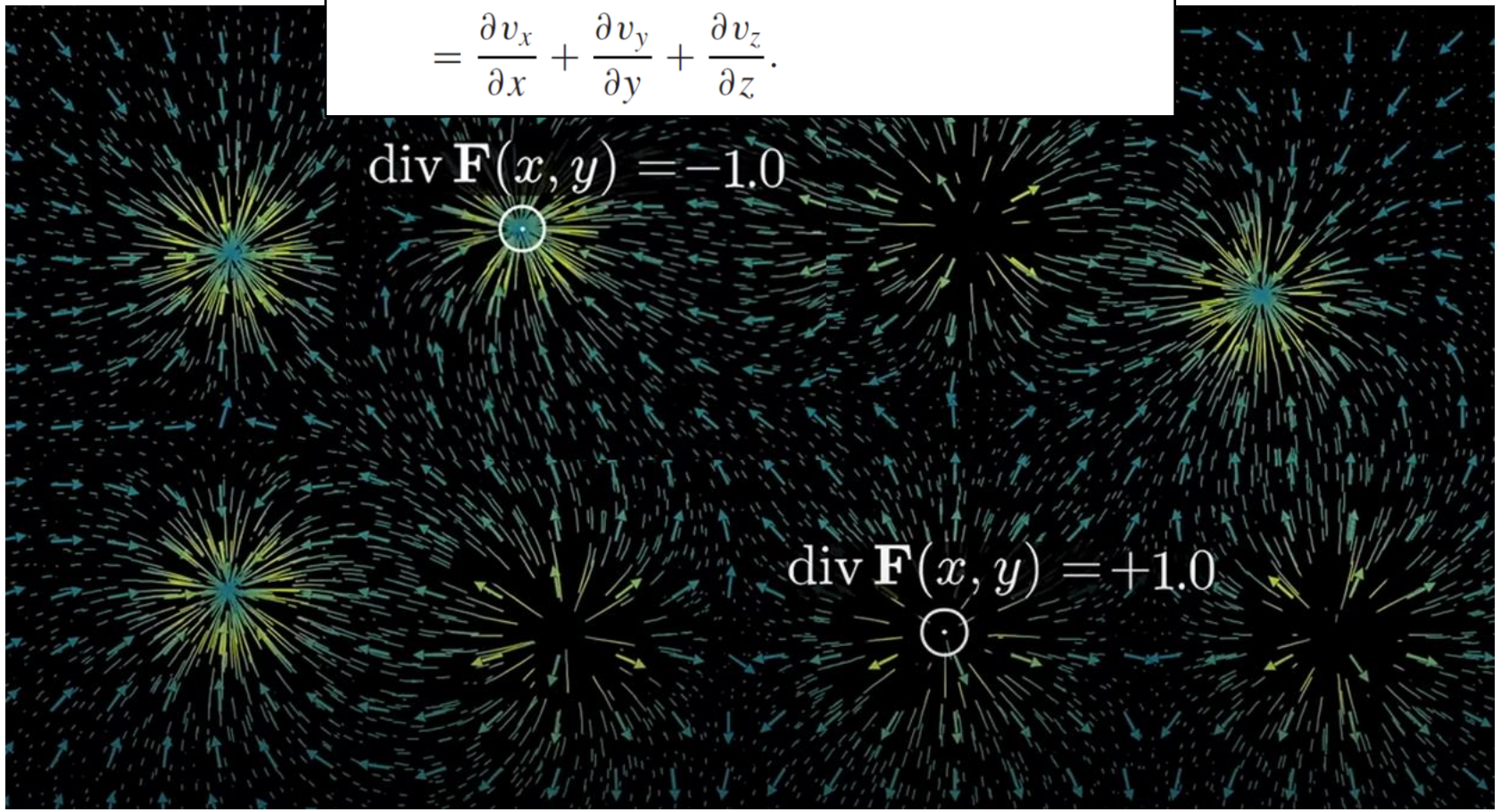
The potential energy of a certain particle is $U = Axy^2 + B \sin Cz$, where A, B and C are constants. What is the corresponding force?

$$\mathbf{F} = -(\hat{\mathbf{x}}\, Ay^2 + \hat{\mathbf{y}}\, 2Axy + \hat{\mathbf{z}}\, BC \cos Cz)$$

∇ OPERATOR

- Divergence

$$\nabla \cdot \mathbf{v} = \left(\hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{z}}\frac{\partial}{\partial z}\right) \cdot (v_x\hat{\mathbf{x}} + v_y\hat{\mathbf{y}} + v_z\hat{\mathbf{z}})$$
$$= \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}.$$



How much the vector spreads out – a scalar quantity

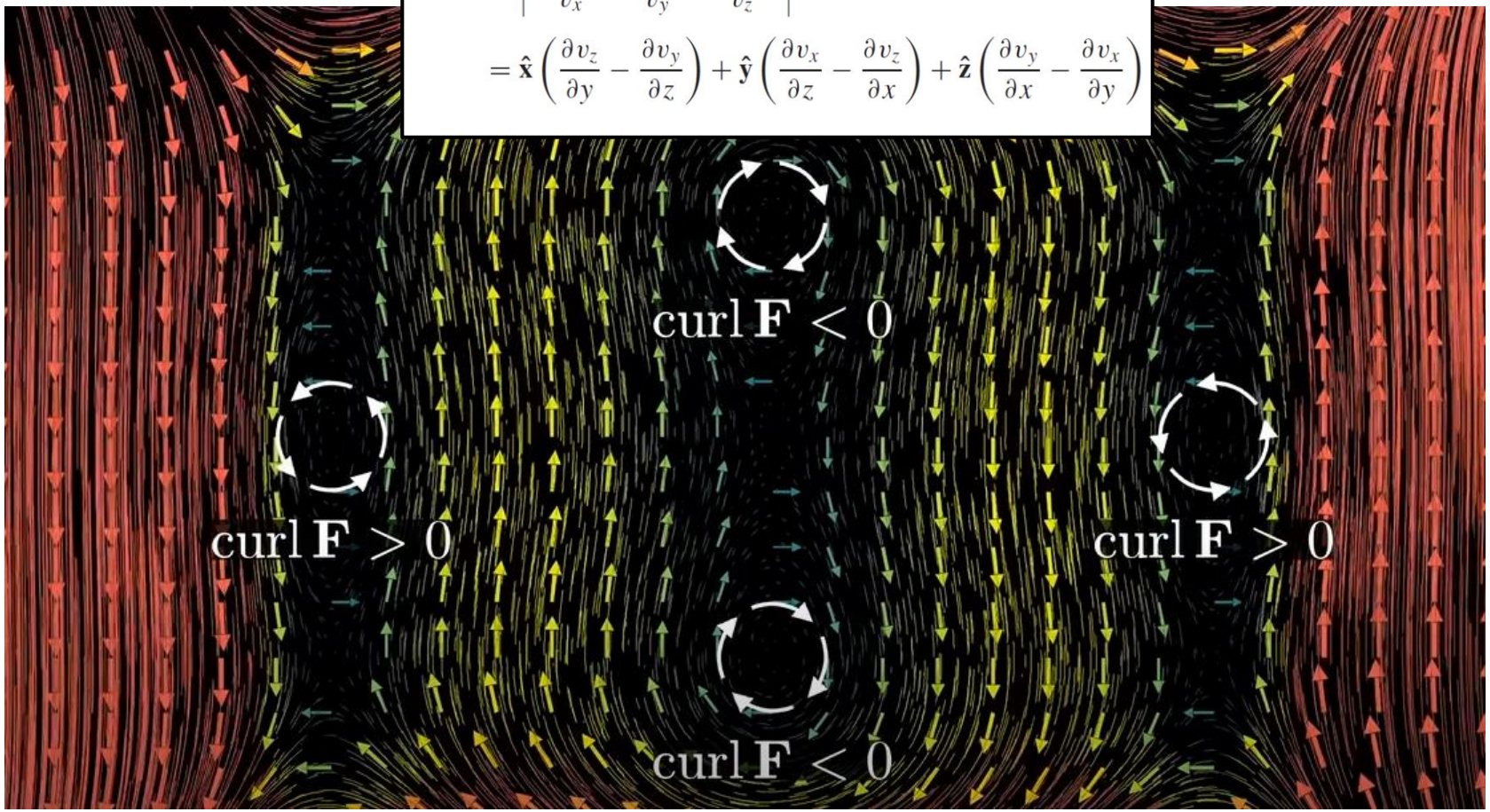
3Blue1Brown
5.01M subscribers

∇ OPERATOR

- Curl

$$\nabla \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ v_x & v_y & v_z \end{vmatrix}$$

$$= \hat{\mathbf{x}}\left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right) + \hat{\mathbf{y}}\left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right) + \hat{\mathbf{z}}\left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)$$

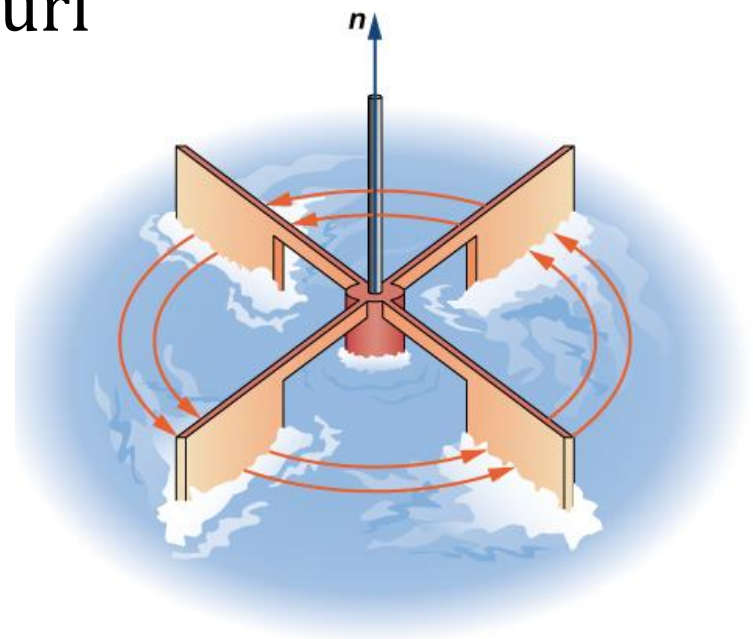


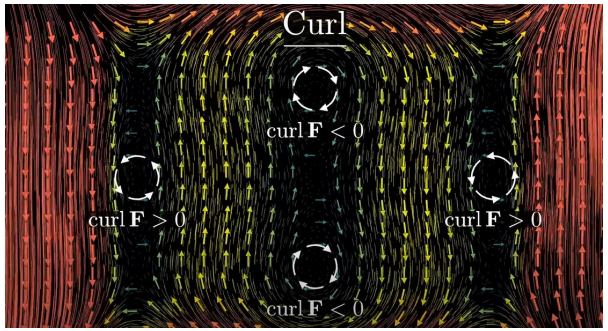
How much the vector curls around – a vector quantity

3Blue1Brown
5.01M subscribers

∇ OPERATOR

- Curl





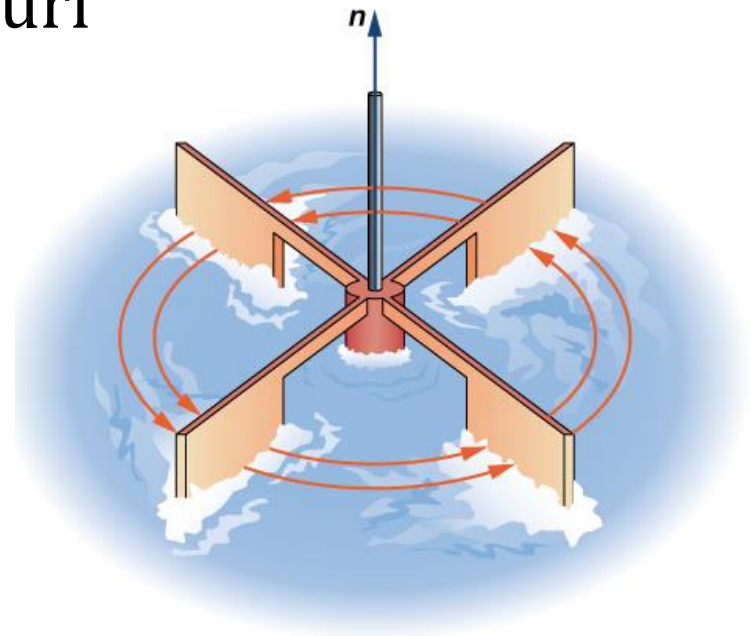
$$\begin{aligned}\operatorname{curl}\vec{\mathbf{F}} &= (R_y - Q_z)\,\hat{\mathbf{i}} + (P_z - R_x)\,\hat{\mathbf{j}} + (Q_x - P_y)\,\hat{\mathbf{k}} \\ &= \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\hat{\mathbf{i}} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\hat{\mathbf{j}} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\hat{\mathbf{k}}.\end{aligned}$$

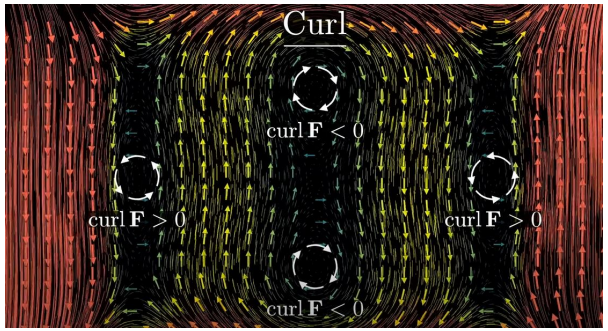
Exercise:

Find the curl of $\vec{\mathbf{F}}(P, Q, R) = \langle x^2 z, e^y + xz, xyz \rangle$

∇ OPERATOR

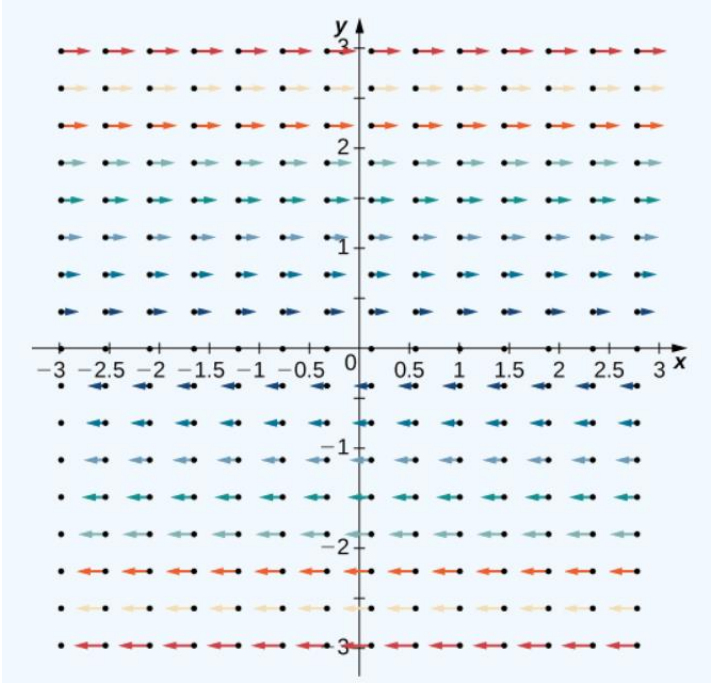
- Curl



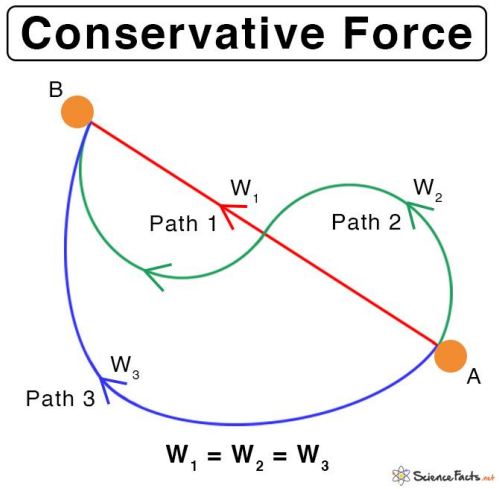


In 2D: $\operatorname{curl}\vec{\mathbf{F}} = (Q_x - P_y)\,\hat{\mathbf{k}} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\hat{\mathbf{k}}.$

Exercise: Find the curl of $\vec{\mathbf{F}} = \langle P, Q\rangle = \langle y, 0\rangle$



CONSERVATIVE FORCE



$\int_1^2 \mathbf{F} \cdot d\mathbf{r}$ is path-independent

Another condition:

Curl of a gradient is zero!

$$\nabla \times \mathbf{F} = 0$$

vector	x component	y component	z component
∇	$\partial/\partial x$	$\partial/\partial y$	$\partial/\partial z$
$\mathbf{F}$	F_x	F_y	F_z
$\nabla \times \mathbf{F}$	$\frac{\partial}{\partial y}F_z - \frac{\partial}{\partial z}F_y$	$\frac{\partial}{\partial z}F_x - \frac{\partial}{\partial x}F_z$	$\frac{\partial}{\partial x}F_y - \frac{\partial}{\partial y}F_x$

CONSERVATIVE FORCE

1) The value of the line integral of a conservative vector field $\boldsymbol{F}(\boldsymbol{r})$ from any point P to any point Q depends only on the end points P and Q and not on the actual path taken from P to Q.

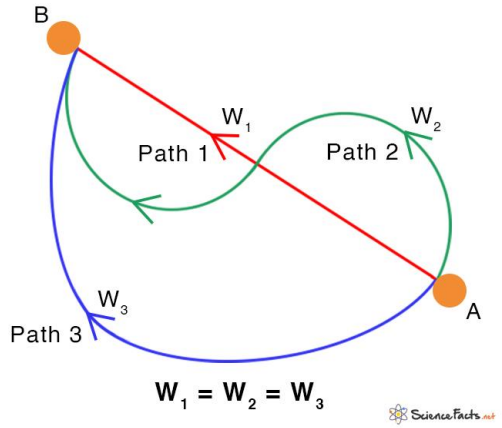
2) The line integral of a conservative vector field $\boldsymbol{F}(\boldsymbol{r})$ around any closed path is equal to zero.

3) A conservative vector field $\boldsymbol{F}(\boldsymbol{r})$ can be written as the gradient of a scalar field,

$$\boldsymbol{F}(\boldsymbol{r}) = -\boldsymbol{\nabla} U(\boldsymbol{r})$$

4) All conservative vector fields are irrotational vector fields,

$$\boldsymbol{\nabla} \times \boldsymbol{F}(\boldsymbol{r}) = -\boldsymbol{\nabla} \times \boldsymbol{\nabla} U(\boldsymbol{r}) = 0$$



CONSERVATIVE FORCE: EXAMPLE

Identifying conservative forces Which of the following forces is/are conservative?

(a) $\boldsymbol{F} = k\,(x, 2y, 3z)$, where k is a constant.

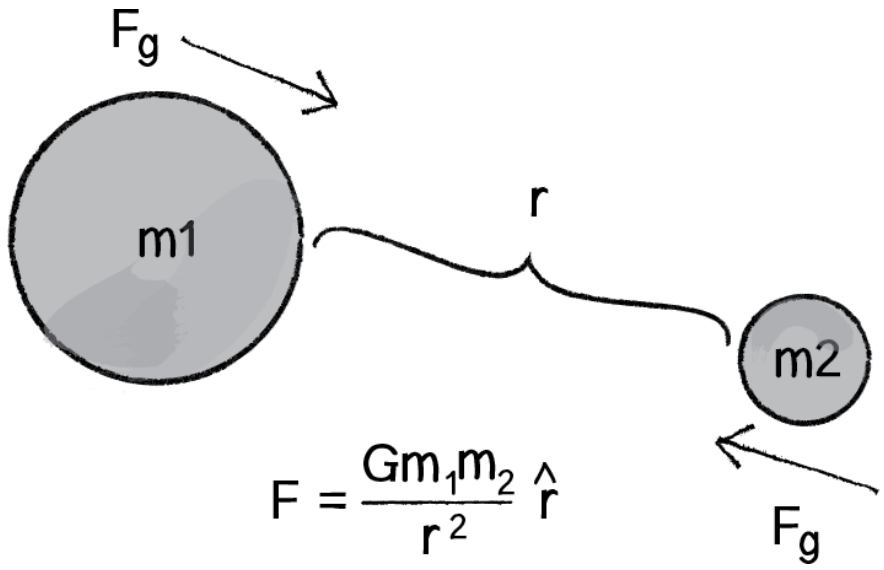
(b) $\boldsymbol{F} = k\,(y, x, 0)$.

(c) $\boldsymbol{F} = k\,(-y, x, 0)$.

For those that are conservative, find the corresponding potential energy U, and verify by direct differentiation that $\boldsymbol{F} = -\boldsymbol{\nabla} U$. [JRT Problem 4.23]

CONSERVATIVE FORCE: EXAMPLE

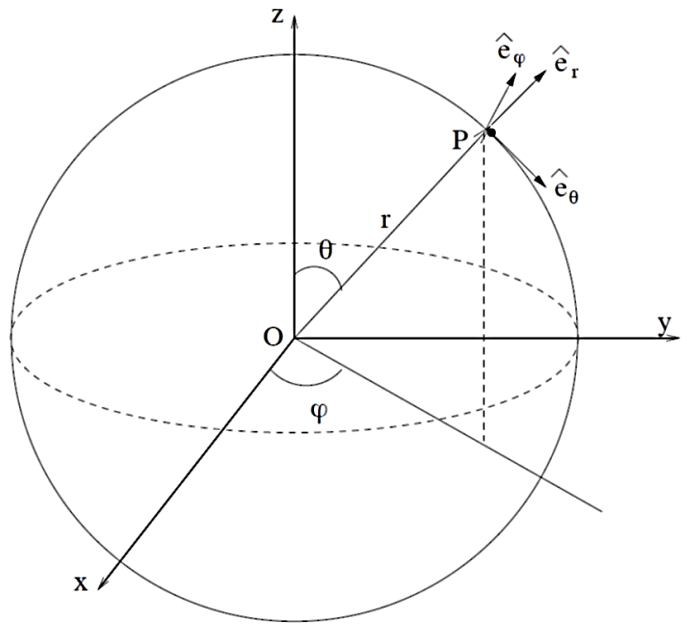
Show that the gravitational force is a conservative force



SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi \quad , \quad y = r\sin\theta\sin\varphi \quad , \quad z = r\cos\theta$$



The range of values of (r, θ, φ) that cover all points in space are:

$$0 \leq r < \infty \quad , \quad 0 \leq \theta \leq \pi \quad , \quad 0 \leq \varphi < 2\pi \ .$$

SPHERICAL COORDINATE SYSTEM

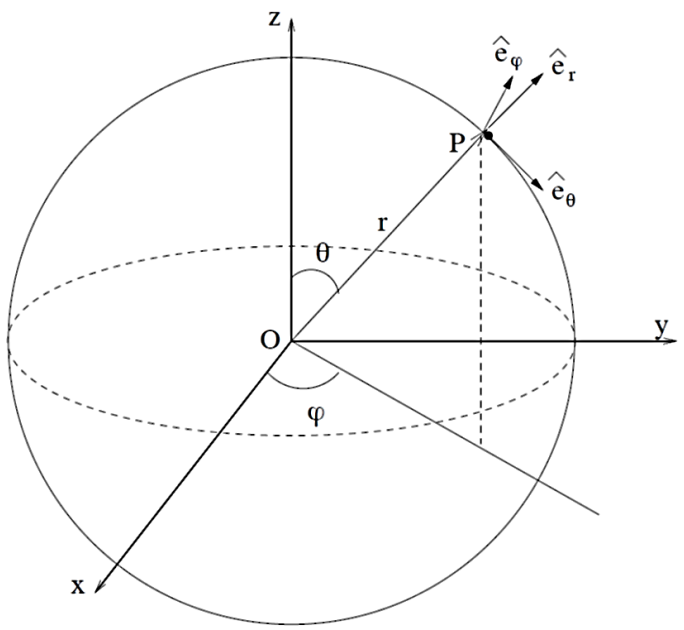
Position vector $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

$$\Rightarrow \boxed{\vec{r} = r\sin\theta\cos\phi\,\hat{x} + r\sin\theta\sin\phi\,\hat{y} + r\cos\theta\,\hat{z}}$$

$$\hat{r} = \frac{\partial\vec{r}/\partial r}{|\partial\vec{r}/\partial r|} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z}$$

$$\hat{\theta} = \frac{\partial\vec{r}/\partial\theta}{|\partial\vec{r}/\partial\theta|} = \cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}$$

$$\hat{\phi} = \frac{\partial\vec{r}/\partial\phi}{|\partial\vec{r}/\partial\phi|} = -\sin\phi\,\hat{x} + \cos\phi\,\hat{y}$$



The range of values of (r, θ, φ) that cover all points in space are:

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SPHERICAL COORDINATE SYSTEM

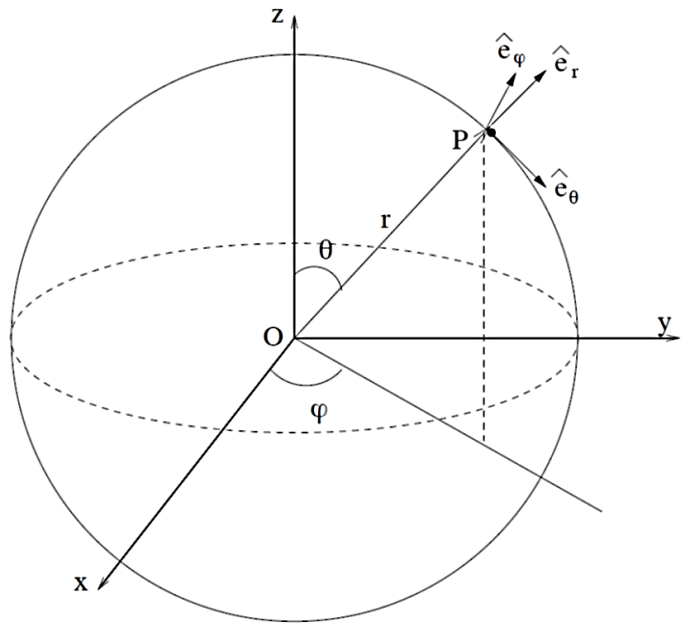
Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi \quad , \quad y = r\sin\theta\sin\varphi \quad , \quad z = r\cos\theta$$

Unit vectors:

$$\begin{aligned}
\hat{e}_r &= \sin\theta\cos\varphi\ \hat{e}_x + \sin\theta\sin\varphi\ \hat{e}_y + \cos\theta\ \hat{e}_z \\
\hat{e}_\theta &= \cos\theta\cos\varphi\ \hat{e}_x + \cos\theta\sin\varphi\ \hat{e}_y - \sin\theta\ \hat{e}_z \\
\hat{e}_\varphi &= -\sin\varphi\ \hat{e}_x + \cos\varphi\ \hat{e}_y \quad .
\end{aligned}$$

$$\hat{e}_r \times \hat{e}_\theta = \hat{e}_\varphi \ , \ \hat{e}_\theta \times \hat{e}_\varphi = \hat{e}_r \text{ and } \hat{e}_\varphi \times \hat{e}_r = \hat{e}_\theta$$



The range of values of (r, θ, φ) that cover all points in space are:

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SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

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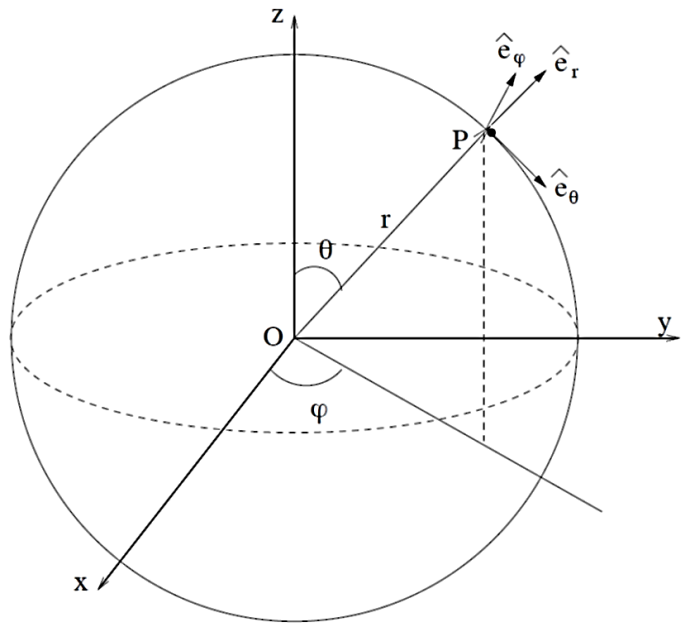
Unit vectors:

$$\begin{aligned}
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\hat{e}_\theta &= \cos\theta\cos\varphi\ \hat{e}_x + \cos\theta\sin\varphi\ \hat{e}_y - \sin\theta\ \hat{e}_z \\
\hat{e}_\varphi &= -\sin\varphi\ \hat{e}_x + \cos\varphi\ \hat{e}_y \quad .
\end{aligned}$$

$$\hat{e}_r \times \hat{e}_\theta = \hat{e}_\varphi \ , \ \hat{e}_\theta \times \hat{e}_\varphi = \hat{e}_r \text{ and } \hat{e}_\varphi \times \hat{e}_r = \hat{e}_\theta$$

Position vector: $\vec{r} = r\ \hat{e}_r$

Line element: $\delta\vec{r} = \delta r\ \hat{e}_r + r\ \delta\theta\ \hat{e}_\theta + r\sin\theta\ \delta\varphi\ \hat{e}_\varphi$



The range of values of (r, θ, φ) that cover all points in space are:

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SPHERICAL COORDINATE SYSTEM

Relations with the cartesian coordinates:

$$x = r\sin\theta\cos\varphi \quad , \quad y = r\sin\theta\sin\varphi \quad , \quad z = r\cos\theta$$

Unit vectors:

$$\begin{aligned}
\hat{e}_r &= \sin\theta\cos\varphi\ \hat{e}_x + \sin\theta\sin\varphi\ \hat{e}_y + \cos\theta\ \hat{e}_z \\
\hat{e}_\theta &= \cos\theta\cos\varphi\ \hat{e}_x + \cos\theta\sin\varphi\ \hat{e}_y - \sin\theta\ \hat{e}_z \\
\hat{e}_\varphi &= -\sin\varphi\ \hat{e}_x + \cos\varphi\ \hat{e}_y \quad .
\end{aligned}$$

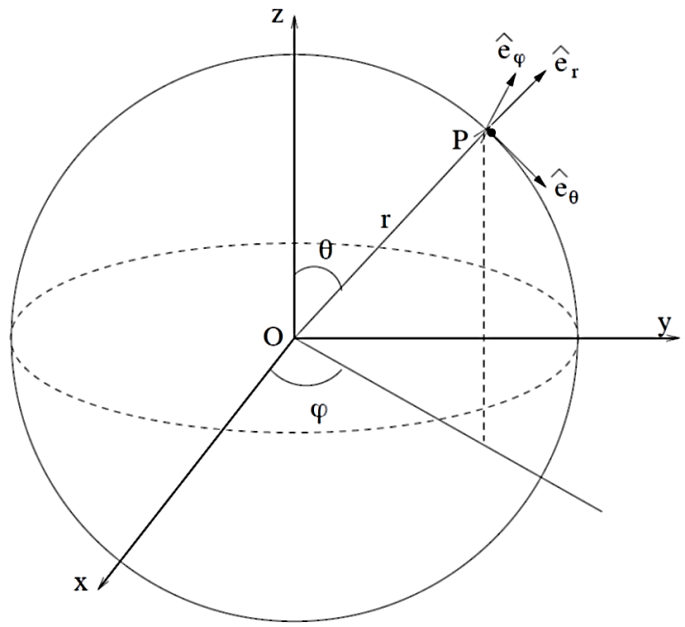
$$\hat{e}_r \times \hat{e}_\theta = \hat{e}_\varphi \ , \ \hat{e}_\theta \times \hat{e}_\varphi = \hat{e}_r \text{ and } \hat{e}_\varphi \times \hat{e}_r = \hat{e}_\theta$$

Position vector: $\vec{r} = r\ \hat{e}_r$

Line element: $\delta\vec{r} = \delta r\ \hat{e}_r + r\ \delta\theta\ \hat{e}_\theta + r\sin\theta\ \delta\varphi\ \hat{e}_\varphi$

Area element: $dS = (R^2\sin\theta\ d\theta\ d\varphi)\ \hat{e}_r$

Volume element: $dV = r^2\sin\theta\ dr\ d\theta\ d\varphi$



The range of values of (r, θ, φ) that cover all points in space are:

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